L-functions and deformations of reduced mod p complex manifolds

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1 Formulating the problem

We study a special class of non-singular complex projective varieties: those that admit a (good) reduction modulo p (prime) via their families parameterized by some "base" variety and, in the spirit of Weil, we associate L functions with them and shall attempt to reduce the problem of deformations of these manifolds to the study of these L functions.
Let $q = p^m$ be a prime power and $\mathbf{F}_q$ be a finite field with q-elements. Then Weil defined the zeta function associated to the variety over this field $X_0/\mathbf{F}_q$ as,

$$Z(X_0/\mathbf{F}_q, t) = exp(\sum_{m \geq 1} \frac{N_m}{m} t^m)$$

where $|X_0/\mathbf{F}_q| := N_m$ (number of points of the variety) and conjectured some remarkable properties of this function.
We focus on one of these properties: If $X(\mathbf{C})$ is a non-singular complex projective manifold of which the reduction modulo p is $X_0/\mathbf{F}_q$, then the betti numbers of $X(\mathbf{C})$ can be recovered from $Z(X_0/\mathbf{F}_q, t)$. More precisely, $Z(X_0/\mathbf{F}_q, t)$ can be written as a ratio

$$\frac{P_1(t)P_3(t)...P_{2n-1}(t)}{P_0(t)P_2(t)...P_{2n}(t)}$$

where the P_i's are integral polynomials. Then,

$$degP_i(t) = dim_{\mathbf{C}} H^i(X(\mathbf{C}), \mathbf{C})$$

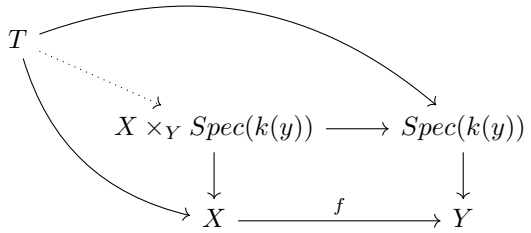
Consider the family $\{X_T(\mathbf{C})\}_{T\in I}$, where each X_T's admit a (good) reduction modulo p, giving us a family $\{X_{0,T}/\mathbf{F}_q\}_{T\in I}$ such that the condition above holds.
We recall that a complex analytic family of compact complex analytic manifolds is given as $\omega : \mathcal{M} \to B$ such that ω is a holomorphic map, for each $T \in B \subset \mathbf{C}^m$, we have that ω^{-1} is compact submanifold of $\mathcal{M}$ and $X_T(\mathbf{C}) = \omega^{-1}(T)$.
As a preliminary definition we know that two (compact) complex manifolds X, Y are called deformations of each other if $\omega^{-1}(T_1) = X$, $\omega^{-1}(T_2) = Y$ for some $T_1, T_2 \in B$. So now we state

Question 1: can we construct a flat morphism of schemes $\omega_p : \mathcal{M}_p \to B'$ such that for some $T_1', T_2' \in B'$ it will hold that $\omega_p^{-1}(T_1') = X_0/\mathbf{F}_q$ and $\omega_p^{-1}(T_2') = X_1/\mathbf{F}_q$ for which the reduction modulo p of $X_0/\mathbf{F}_q$ and $X_1/\mathbf{F}_q$ are precisely X, Y (respectively) that are mentioned above to be deformations of each other?

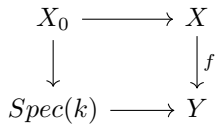
Since the betti numbers are topological invariants, the function $Z(X_0/\mathbf{F}_q, t)$ doesn't "care" at all about deforming the complex structure. We are thus motivated to construct a zeta function attached to the reduced modulo p projective non-singular varieties $X_0/\mathbf{F}_q$ (whose complex non-singular version is $X(\mathbf{C})$) that closely depends on the complex structure of $X(\mathbf{C})$.

2 Deforming the variety

We recall the algebraic definition of deformations (namely those of schemes) (chapter 2 [2]). It is known that the fibre $X_y = f^{-1}(y)$ is topologically the fibre product



where $y \in Y$ is a point and $Spec(k(y))$ is the residue field, for a morphism $f : X \to Y$. We see that X_y is a scheme over $Spec(k(y))$. Take the scheme X_0 over the field k, we define a family of deformations of X_0 to be a morphism $f : X \to Y$ with connected Y together with a point $y_0 \in Y$ such that $k(y_0) = k$ and $X_{y_0} \cong X_0$ and the other fibres X_y are the deformations of X_0. These deformations can be shown by the pullback square as follows



For the scheme X_0 over the finite field $\mathbf{F}_q$ we can take the following example of pullback square as its (arithmetic) deformations

$$\begin{array}{ccc} X_0 & \longrightarrow & \mathcal{M}_p \\ \downarrow & & \downarrow{\scriptstyle \omega_p} \\ Spec(\mathbf{F_q}) & \longrightarrow & Spec(\mathbf{Z}/(p^2)) \end{array}$$

to which we want to compare the diagram

$$\begin{array}{ccc} X(\mathbf{C}) & \longrightarrow & \mathcal{M} \\ \downarrow & & \downarrow{\scriptstyle \omega} \\ \{t_0\} & \longrightarrow & B \end{array}$$

Such that X_0 over $\mathbf{F}_q$ is the reduction modulo p of $X(\mathbf{C})$. Now for a scheme X over $Spec(\mathbf{Z})$ of finite type (whose underlying topological space is $|X|$) the zeta function associated with the scheme is [1]

$$\zeta(X, s) = \prod_{x \in |X|} \frac{1}{1 - (\#k(x))^{-s}}$$

We want to extend this to the Weil zeta functions of the fibres X_y of X_0 over the field k, which are the deformations of X_0.More precisely, keeping Question 1 (from the previous section) in mind, we study schemes X_0 over finite fields $\mathbf{F}_q$ which are reduction modulo p of some complex manifolds $X(\mathbf{C})$.

References

[1] Pierre Deligne. The weil conjectures i. *arXiv preprint arXiv:1807.10810*, 2018.

[2] Robin Hartshorne. *Algebraic geometry*, volume 52. Springer Science & Business Media, 2013.